

The basic issue: Population growth and the quality of life

World Population in 2009 6.8 Billions

Projected Population in 2050 9.2 Billions

- What will happen to development if such projections are realized?
- Is this scenario inevitable or will it depend upon success or failure of our development efforts?
- Is population growth per se a problem or it is a manifestation of underdevelopment and unequal utilization of global resources?

Every year more than 75 million people are being added to world population

Almost all the increase is in developing countries (97%)

Population growth is not a problem of numbers, it is a problem of development

How does Population affect development?

How does development affect population growth?

- Will developing countries be capable of improving the levels of living?
- How will the developing countries be able to cope with the vast increase in their labor force?
- Will world food supply and its distribution be sufficient?
- Will developing countries be able to extend health and educational system?
- Is there a relationship between poverty and family size?
- Is inexorable pursuit of increasing affluence among the rich more detrimental to living standards among the poor?

Population growth: Past, Present and Future

In absolute numbers

12,000 years ago world population was just 5 Millions
2,000 years ago (year 1) 250 Millions (less than 20 % of China today)
From year 1 to 1750, it tripled to 728 Millions (less than 75 % of India today)
In next 200 year (1750-1950) 1.7 Billions peoples were added
In next four decades (1950-1990) more than doubled to 5.3 Billions
The world entered the 21st century with 6 Billion
Projected population in 2050 9.2 Billions

In percentage

From beginning to 1700 the population increased at Zero percent (.002%)
By 1750 the growth rate accelerated to .3 %
By 1950 the growth rate tripled to 1 %
In 1970 it peaked at 2.35%
Today it is almost 1.1 % but the rate of increase is slowing (still 2.3 % in Africa)

Doubling time

Before 1650 it took 36,000 year or 1,400 generations to double
Today it takes 58 years or 2 generation to double
Year 1- 1750 it took 1750 years to add 480 Millions peoples and today same number is being added in less than 7 years

TABLE 6.1 Estimated World Population Growth

Year	Estimated Population (millions)	Estimated Annual Increase in the Intervening Period (%)	Doubling time (years)
10,000 B.C.E.	5		
1 C.E.	250	0.04	1,733
1650	545	0.04	1,733
1750	728	0.29	239
1800	906	0.45	154
1850	1,171	0.53	130
1900	1,608	0.65	106
1950	2,576	0.91	76
1970	3,698	2.09	33
1980	4,448	1.76	39
1990	5,292	1.73	40
2000	6,090	1.48	47
2010	6,892	1.22	57
2050 (projected)	9,200	0.675	103

Sources: Population Reference Bureau, *World Population Data Sheet* (Washington, D.C.: Population Reference Bureau, 2010 and previous annuals). Warren S. Thompson and David T. Lewis, *Population Problems*, 5th ed. (New York: McGraw-Hill, 1965), p. 384; United Nations, *Demographic Yearbook for 1971* (New York: United Nations, 1971); United Nations, *Report on the World Social Situation, 1997* (New York: United Nations, 1997), p. 14, and authors' calculations. An alternate system of broadly comparable and earlier estimates is found in Michael Kremer, "Population Growth and Technological Change: One Million B.C. to 1990," *Quarterly Journal of Economics* 108 (1993): 681-716.

Demographic transition: Western Europe

Definition

The process by which fertility rates eventually declines to replacement level.

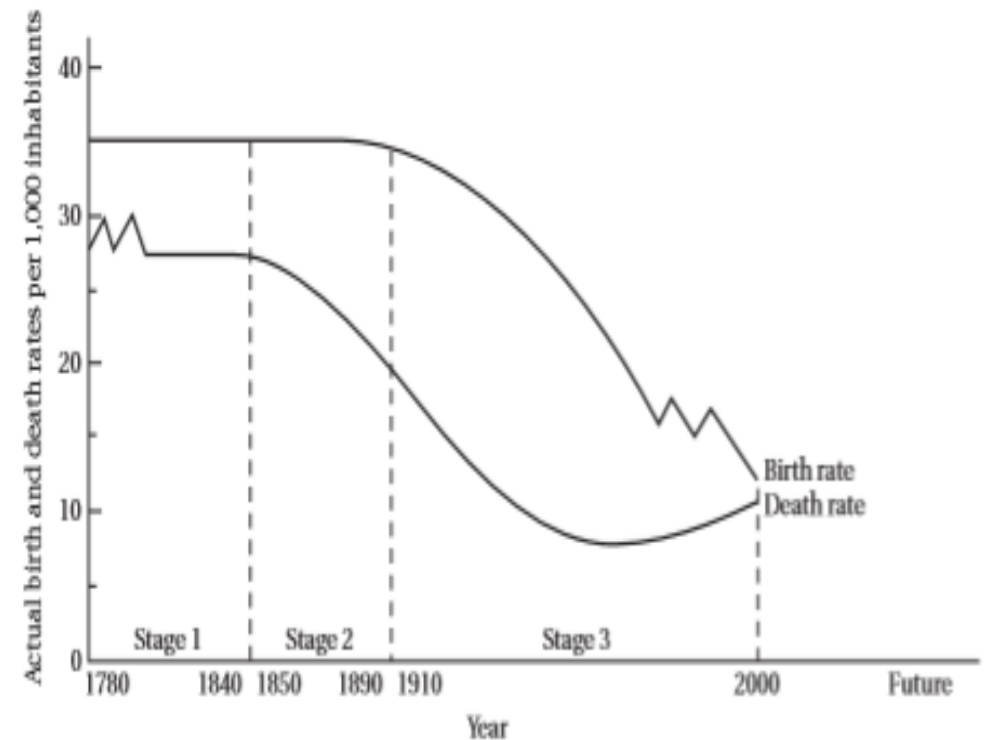
The phasing-out process of population growth rates from a virtually **stagnant growth stage characterized by high birth rates and death rates** through a **rapid-growth stage with high birth rates and low death rates** to a stable, **low growth stage in which both birth and death rates are low**.

Stage I: High birth rate and high death rate
Birth rate hovered around 35 per 1,000
Death rate fluctuated around 40 per 1,000
Low population growth rate (0.5%)
High stationary state

Stage II: Mortality rate declines due to: Modernization, better public health methods, healthier diet, higher income
life expectancy rises from 40 to 60
High birth rate
Rapid population growth (1%)

Stage III: Modernization and development caused decline in fertility
Low mortality and low birth rate
Slow or no population growth

FIGURE 6.5 The Demographic Transition in Western Europe



Demographic transition: Developing countries

Stage I

High birth rate and high death rate

Both the rate are higher than preindustrial Europe

Low population growth

Stage II

Death rate fell more rapidly than pre industrial Europe

Given high birth rate (40 per 1,000) population growth rate remained over 2% (more than western Europe 1%)

Stage III

Class A (Taiwan and Korea etc)

Due to modern death control methods and better level of living death rate fell to 10 per 1,000 and birth rate to 25 per 1,000

Rapidly falling rates of population growth

Class B (Africa and Middle east)

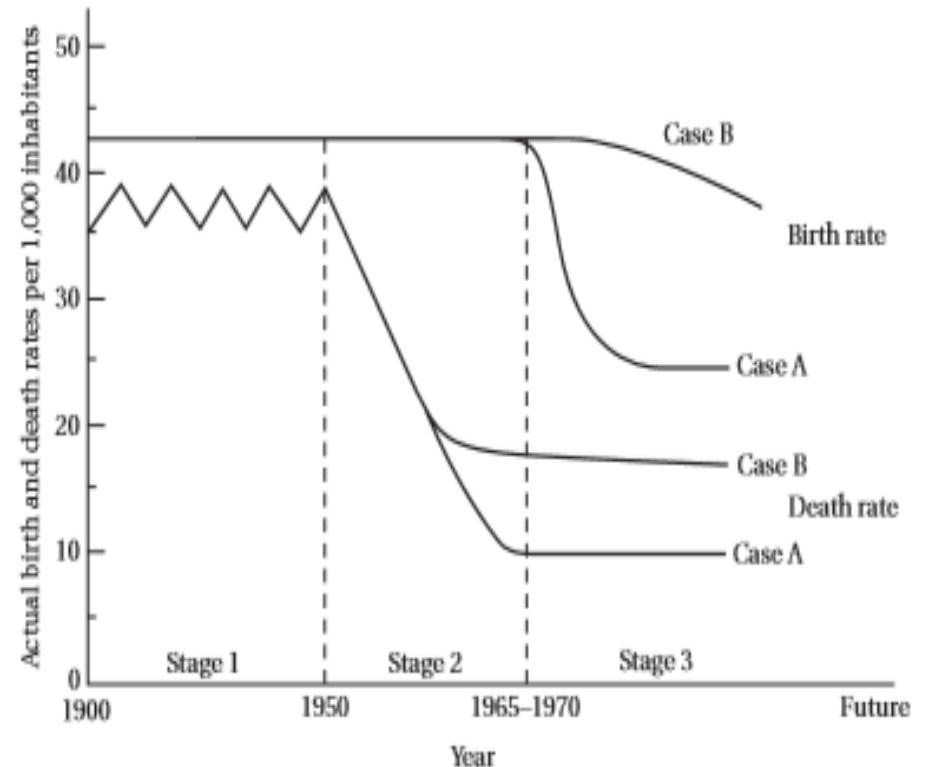
Birth rate remained high due to low level of living

Death rate failed to drop further due to wide spread poverty, low level of living and epidemics like AIDS

High growth rate of population

These are still in stage II

FIGURE 6.6 The Demographic Transition in Developing Countries



Source: Adapted from National Academy of Sciences, *The Growth of World Population* (Washington, D.C.: National Academy of Sciences, 1963), p. 15.



Consequences of high fertility: Some conflicting perspectives

It's not a real problem

The problem is Other issues

Underdevelopment

If correct strategies are pursued and lead to higher levels of living, population will take care of itself.
Underdevelopment is the real problem.

World resource depletion and environmental destruction

Issue is population in relation to world resources

Less than 25% population of developed countries consume 80% resources.

Affluent nations should reduce their high consumption instead of asking LDCs to reduce their population

Population distribution

Problem is not number of peoples, but their distribution in space

Some regions are underpopulated and some are overpopulated

Subordination of women

Women's inferior role, low status, and restricted access to birth control are manifested in high fertility

Their education, health and economic well being should be improved

It's a deliberately contrived false issue

This point is allied to Neo-classical dependence theory

Rich nations intends to maintain a status quo by holding down the population of poor nations

Neo-Marxist view is: These are genocidal attempts by rich countries and their allied agencies

Developed countries consider population growth a threat to their welfare

Efforts of developed countries to control population is manifestation of their fear

It's a desirable phenomenon

Large population provide the needed consumer demand to generate favorable economies of scale in production, to lower production costs, and to provide a sufficient low cost labor supply to achieve higher output levels,

Neo-classical counter revolutionists say Free market and human ingenuity will solve all the problems arising from population growth.

In many underpopulated countries, much unused but arable land could yield large increases in agricultural output, for example Congo

Ratio of population to arable land. South of sahara has 1.4 Billion hectors arable land but only 12% under cultivation.

Large family size of minorities can be justified

Military and political power depends on large population

It is a real problem

The extremist argument: population and global crisis

Population is the principal cause of poverty, low levels of living, malnutrition, ill health, environmental degradation, and other problems. Coercive measures may be adopted to control population bomb.

The theoretical argument: population-poverty cycle

Poverty and high population growth reinforce each other

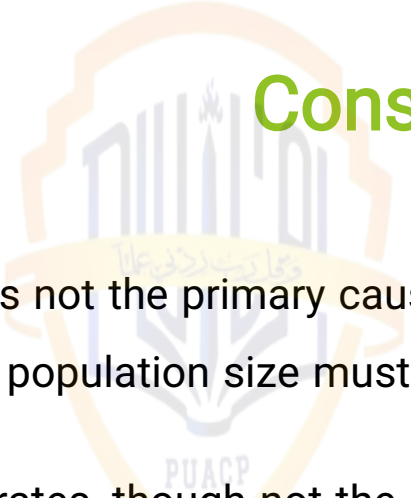
Population intensifies the underdevelopment

It retards the prospects of better life for the already born by lowering the saving at household and national level

It severely draws down Govt. revenues to provide better services to the additional people

Other empirical arguments: seven negative consequences

1. Economic growth
2. Poverty and inequality
3. Education
4. Health
5. Food
6. Environment
7. International migration



Consensus

Population growth is not the primary cause of low levels of living

Developing country population size must be viewed in conjunction with developed country affluence

Population growth rates, though not the principal cause of underdevelopment, are nevertheless important contributing factor



The causes of high fertility in developing countries

The Malthusian population trap

Macroeconomic theory

Theory of relationship between population growth and economic development (per capita income)

Population grows at a geometric rate unless checked by dwindling food supplies, doubling every 30 or 40 years

Food supplies grow only at arithmetic rate due to diminishing returns

Food supplies can not keep pace with burgeoning population and lead to stable population at subsistence level

The only way to avoid this absolute poverty is to limit the population

Thomas Malthus is father of birth control movement

Modern Economists call it Malthusian population trap

This trap can be illustrated with two curves, Population growth rate and aggregate income growth rate

Population growth curve

When the income is very low there is starvation, population does not grow.(demographic transition)

When income rises after that threshold, population grows and eventually reaches the peak (stage II)

Ultimately falls to a stable population (zero growth rate-stage III)

This pattern is in accordance with demographic transition

Total income growth curve

Positive relationship between total income growth and development (per capita income)

Eventually growth levels off at a maximum

Per capita Income growth

Difference between income growth and population growth is per capita income growth

Growth rate of income > Growth rate of population.....Per Capita Income rises

Growth rate of income < Growth rate of population.....Per Capita Income falls

Growth rate of income = Growth rate of population.....Per Capita Income unchanged

Trap

The threshold population level anticipated by Malthus at which population increase was bound to stop because life-sustaining resources, which increase at an arithmetic rate, would be insufficient to support human population, which increases at a geometric rate.

The two curves cross at S, it is stable equilibrium, when per capita income increases, population growth is greater than income growth, per capita income fall back to S.

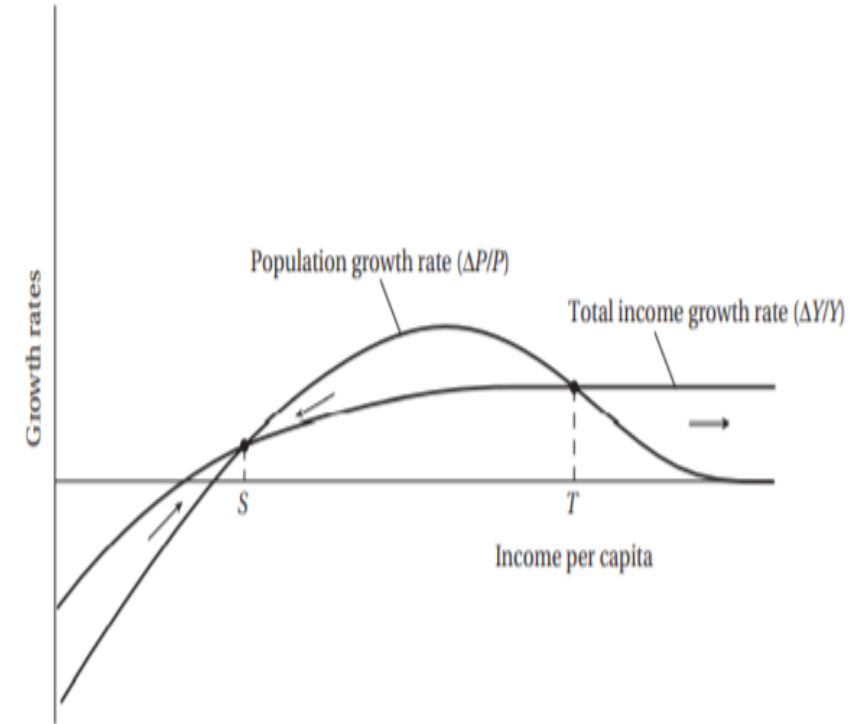
If per capita income were a little less than S, population growth is less than income growth, per capita income rises back to S.

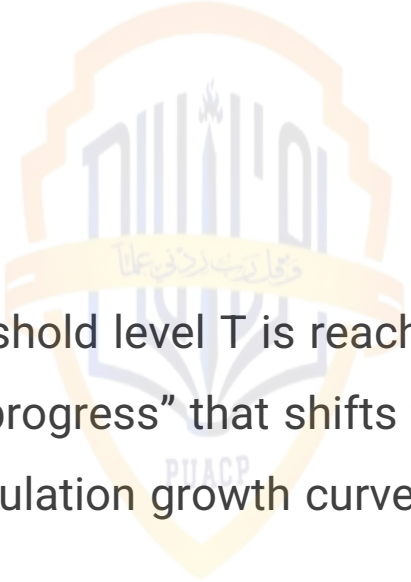
This is a low level equilibrium, Malthusian population trap

Poor countries can not escape unless they initiate preventive checks (birth control) on population growth

Otherwise the Malthusian positive checks (starvation etc) will check the population growth

FIGURE 6.7 The Malthusian Population Trap





Escape

If somehow a threshold level T is reached, per capita income will grow continually

By “technological progress” that shifts the income growth curve upward

By shifting the population growth curve downward by “social progress”

Limited relevance for developing countries like Pakistan

1. They do not take adequate account of the role and impact of technological progress.
2. Macro relationship between population growth and levels of per capita income does not hold.
3. They focus on the wrong variable, per capita income, rather the principle of population growth is family income.

Microeconomic household theory of fertility

Definition:

The theory that family formation has **costs and benefits** that determine the size of families formed

A consumer tries to maximize satisfaction subject to his income and relative prices of goods

Children is a good, fertility is the response to demand for children

Demand for children depends upon household income, price (cost) of children, taste for other goods relative to children and price of other goods

$$C_d = f(Y, P_c, P_x, t_x)$$

C_d = demand for surviving children

Y = Household income

P_c = Net price of children (difference between anticipated cost and benefits)

P_x = the price of all other goods

t_x = the taste for goods relative to children

$\partial C_d / \partial Y > 0$ demand for children varies directly with household income

$\partial C_d / \partial P_c < 0$ It varies inversely with net price of children

$\partial C_d / \partial P_x > 0$ It varies directly with price of goods

$\partial C_d / \partial t_x < 0$ It varies inversely with taste for goods

Household's desires for children are expressed in terms of IC

The ability to purchase is shown with budget constraint

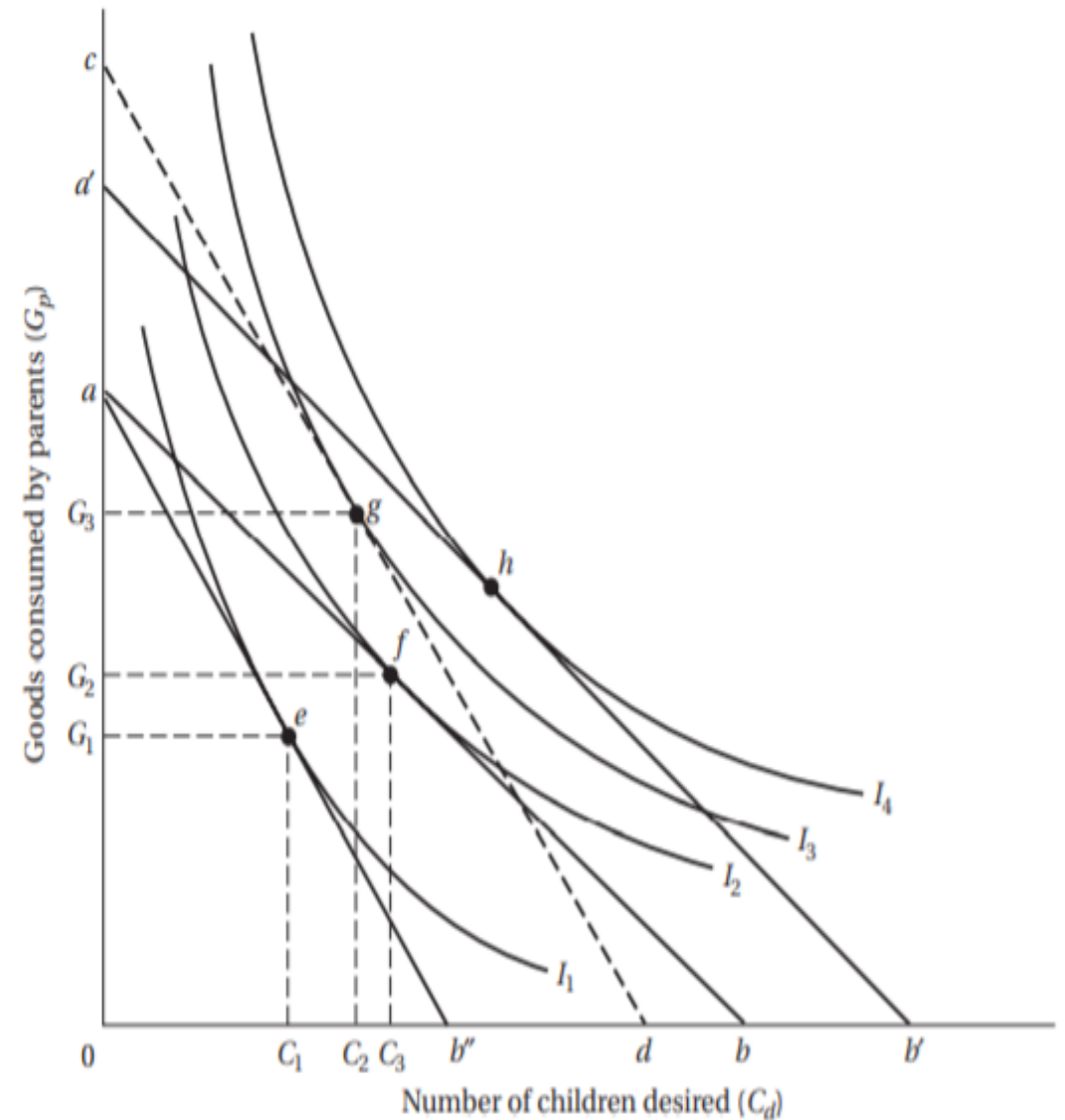
According to this theory, the household chooses that combination which maximizes his family satisfaction

As family income rises, he attains higher satisfaction by consuming more of both goods (point h)

An increase in the price of children causes the household to have fewer children (point e)

A simultaneous increase in income and price of children causes to have fewer children and increase in welfare (point g)

FIGURE 6.9 Microeconomic Theory of Fertility: An Illustration



The demand for children in developing countries

The demand for children depends upon family preference of surviving children, opportunity cost of rearing children, family income

There are psychological and cultural determinants of family size

The choice is with regard to additional (marginal) children

By comparing private cost and private benefits

The theory concludes that when the price of children rises families demand fewer children

One way to induce families to have fewer children is to raise the price of children by providing greater educational opportunities and wide range of higher paying jobs for women